# Summary of How TB Learning Refined to be Neyman Orthogonal

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**Lemma 1** (Non-orthogonality of the density-based moment)**.** *Let $\eta_{1,k}(z,x,w) := p(A = a_k \mid z,x,w)$ denote the conditional density nuisance evaluated at anchor $a_k$, and define the density-based moment condition*

$$M_k^{(1)}(\gamma,\eta_1) := \mathbb{E}\Big[\eta_{1,k}(Z,X,W)\, q_\gamma(Z,a_k,X)\, \hat{h}(W,a_k,X)\Big] - \mathbb{E}\Big[\hat{h}(W,a_k,X)\Big], \tag{1}$$

*where $\hat{h}(W,a_k,X)$ is the outcome bridge evaluated at anchor $a_k$, arising from the proximal Fredholm structure. Then $M_k^{(1)}$ is* not *Neyman orthogonal with respect to $\eta_{1,k}$: for any perturbation direction $\delta\eta_1 \in L^2(P_{Z,X,W})$, the Gateaux derivative at the true values $(\gamma_0, \eta_{1,k}^0)$ satisfies*

$$\mathcal{D}_{\eta_{1,k}}\, \mathbb{E}\Big[M_k^{(1)}(\gamma_0,\eta_{1,k}^0)\Big]\,[\delta\eta_1] = \mathbb{E}\Big[\delta\eta_1(Z,X,W)\cdot q_0(Z,a_k,X)\cdot \hat{h}(W,a_k,X)\Big] \neq 0. \tag{2}$$

*Consequently, estimation error in $\eta_{1,k}$ propagates at first order into the moment condition, and standard semiparametric efficiency arguments do not apply.*

*Proof.* Differentiate $M_k^{(1)}(\gamma_0,\, \eta_{1,k}^0 + \epsilon\,\delta\eta_1)$ with respect to $\epsilon$ at $\epsilon = 0$. The result follows since $q_0 \cdot \hat{h}$ is not identically zero under the identification conditions of the proximal framework. □

**Theorem 2** (Exact Neyman Orthogonality via Fredholm-Structured Correction)**.** *Let $\{a_k\}_{k=1}^K$ be anchor points covering the support of $A$, with anchor-specific bandwidths $\{\alpha_k\}_{k=1}^K$ where $\alpha_k > 0$. Let $K_{\alpha_k}(u) = \frac{1}{\alpha_k}K(u/\alpha_k)$ be a mollifier kernel satisfying*

$$\int K(u)\,du = 1, \qquad \int u\,K(u)\,du = 0, \qquad \kappa_2 := \int u^2\,K(u)\,du < \infty. \tag{3}$$

*Define the correction term motivated by the Fredholm structure of the treatment bridge equation:*

$$\phi_k^{(1)}(D;\gamma,\eta_1) := q_\gamma(Z,a_k,X)\,\hat{h}(W,a_k,X)\cdot\left[K_{\alpha_k}(A-a_k) - \eta_{1,k}(Z,X,W)\right]. \tag{4}$$

*Although $\phi_k^{(1)}$ is defined in terms of $\eta_{1,k}$, its role is purely constructive: in the sum $M_k^{(1)} + \phi_k^{(1)}$, all dependence on $\eta_{1,k}$ cancels algebraically, so the final estimating equation requires no knowledge of $\eta_{1,k}$. The orthogonalized score $\psi_k^{(1)} := M_k^{(1)} + \phi_k^{(1)}$ evaluates explicitly to*

$$\psi_k^{(1)}(D;\gamma) = q_\gamma(Z,a_k,X)\,\hat{h}(W,a_k,X)\,K_{\alpha_k}(A-a_k) - \hat{h}(W,a_k,X). \tag{5}$$

*Then the following hold.*

***(i) Exact Neyman orthogonality.*** *For all $\delta\eta_1 \in L^2(P_{Z,X,W})$:*

$$\mathcal{D}_{\eta_{1,k}}\, \mathbb{E}\Big[\psi_k^{(1)}(D;\gamma_0)\Big]\,[\delta\eta_1] = 0. \tag{6}$$

*We note that once $\psi_k^{(1)}$ is written in the form (5), the NO property holds by inspection, since $\eta_{1,k}$ does not appear. The content of this result is that this form arises from a principled correction of the density-based moment $M_k^{(1)}$ via the Fredholm structure, and that the resulting estimating equation thereby inherits the robustness guarantees of Neyman orthogonal scores: nuisance estimation error in $\eta_{1,k}$ does not propagate into $\hat{\gamma}$ at first order.*

***(ii) Algebraic elimination of the density nuisance.*** *The nuisance $\eta_{1,k} = p(A = a_k \mid Z, X, W)$ is absent from $\psi_k^{(1)}$. Orthogonality is achieved not by debiasing in the sense of standard double machine learning, but by exact algebraic cancellation: the terms $\pm\,\eta_{1,k}\, q_\gamma\, \hat{h}$ cancel identically in the sum $M_k^{(1)} + \phi_k^{(1)}$, leaving only the directly observable quantity $K_{\alpha_k}(A - a_k)$. Estimation error in $\eta_{1,k}$ therefore has no effect on $\psi_k^{(1)}$, regardless of the quality of any density model.*

***(iii) Residual moment bias: smoothing error, not nuisance error.*** *At the true bridge $\gamma_0$, the tower property and a second-order Taylor expansion of $p(a_k + \alpha_k u \mid Z, X, W)$ under the kernel symmetry conditions yield:*

$$\mathbb{E}\Big[\psi_k^{(1)}(D;\gamma_0)\Big] = \underbrace{\frac{\alpha_k^2\,\kappa_2}{2}\;\mathbb{E}\Big[\partial_a^2 p(a_k \mid Z, X, W)\, q_0(Z, a_k, X)\, \hat{h}(W, a_k, X)\Big]}_{\text{leading bias: explicit and estimable}} + O(\alpha_k^4). \tag{7}$$

*The leading term has an explicit structural form determined by the curvature of the conditional treatment density at $a_k$. The remainder $O(\alpha_k^4)$ is the Taylor approximation error in characterizing this bias, not a second independent source of bias. Both terms depend only on the analyst-controlled bandwidth $\alpha_k$ and the local geometry of $p(A \mid Z, X, W)$ near $a_k$; neither depends on estimation error in the density nuisance.*

***(iv) Anchor-local bias-variance optimality of the moment.*** *The mean squared error of the $k$-th kernel-smoothed moment decomposes as*

$$\mathrm{MSE}_k(\alpha_k) = C_k^2\,\alpha_k^4 + \frac{V_k}{N\alpha_k} + O(\alpha_k^6), \tag{8}$$

*where*

$$C_k := \frac{\kappa_2}{2}\;\mathbb{E}\Big[\partial_a^2 p(a_k \mid Z, X, W)\, q_0(Z, a_k, X)\, \hat{h}(W, a_k, X)\Big], \tag{9}$$

$$V_k := R(K)\;\mathbb{E}\Big[q_0^2(Z, a_k, X)\, \hat{h}^2(W, a_k, X)\, p(a_k \mid Z, X, W)\Big], \tag{10}$$

*with $R(K) = \int K(u)^2\,du$. The MSE of the $k$-th moment is minimized by the anchor-specific bandwidth*

$$\alpha_k^* = \left(\frac{V_k}{4C_k^2\,N}\right)^{1/5}, \tag{11}$$

*yielding the local minimax rate $\mathrm{MSE}_k^* = O(N^{-4/5})$ for the moment. What is non-standard here relative to classical kernel smoothing is that $V_k$ depends on the local marginal treatment density $\bar{p}(a_k)$ through the anchor structure, so $\alpha_k^*$ varies across anchors in proportion to $(N\bar{p}(a_k))^{-1/5}$. This per-anchor calibration achieves the local minimax rate uniformly across anchors even when the treatment distribution is nonuniform, a property not available under a global bandwidth.*

*Proof sketch. Part (i).* Perturb $\eta_{1,k} \to \eta_{1,k} + \epsilon\,\delta\eta_1$ in $\psi_k^{(1)}$ and differentiate at $\epsilon = 0$. This gives $\mathbb{E}[\delta\eta_1 \cdot q_0\hat{h}] - \mathbb{E}[\delta\eta_1 \cdot q_0\hat{h}] = 0$, where the two terms arise from $M_k^{(1)}$ and $\phi_k^{(1)}$ respectively.

*Part (ii).* Direct inspection of $M_k^{(1)} + \phi_k^{(1)}$ shows that the terms $\pm\eta_{1,k}\, q_\gamma\, \hat{h}$ cancel algebraically, yielding (5).

*Part (iii).* The tower property gives $\mathbb{E}[K_{\alpha_k}(A-a_k) \mid Z, X, W] = \int K_{\alpha_k}(a-a_k)\, p(a \mid Z, X, W)\, da$. Substituting $u = (a-a_k)/\alpha_k$ and Taylor expanding $p(a_k + \alpha_k u \mid Z, X, W)$ around $a_k$ yields the stated expression; the $O(\alpha_k)$ term vanishes by kernel symmetry, and $O(\alpha_k^4)$ is the third-order Taylor remainder.

*Part (iv).* Minimize $C_k^2\alpha_k^4 + V_k/(N\alpha_k)$ with respect to $\alpha_k$, setting the derivative to zero and solving for $\alpha_k^*$ as in (11). □

A similar analysis holds for the normalization condition, the first term in Eq. 16 of the submitted paper (omitted for brevity); the same algebraic cancellation applies with $\hat{h} \equiv 1$, yielding $\psi_k^{(2)}(D;\gamma) = q_\gamma(Z, a_k, X)\, K_{\alpha_k}(A-a_k) - 1$, which is likewise free of $\eta_{1,k}$.

**Corollary 3** ($\sqrt{N}$-consistency of the bridge estimator under undersmoothing)**.** *Suppose $\alpha_k \to 0$ and $\sqrt{N}\,\alpha_k^2 \to 0$ as $N \to \infty$ (undersmoothing). Then the moment bias* (7) *satisfies $\sqrt{N}\ \mathbb{E}[\psi_k^{(1)}(D;\gamma_0)] \to 0$, so the bias is asymptotically negligible relative to the $N^{-1/2}$ stochastic fluctuations of the empirical moment. Under standard regularity conditions for M-estimation and the identification assumptions of the proximal framework, the bridge estimator $\hat{\gamma}$ solving the stacked moment conditions satisfies*

$$\sqrt{N}\,(\hat{\gamma} - \gamma_0) \ \xrightarrow{d}\ \mathcal{N}(0,\ J^{-1}\,\Sigma\,(J^{-1})^\top), \tag{12}$$

*where $J = \nabla_\gamma\, \mathbb{E}[\psi_k^{(1)}(D;\gamma_0)]$ is the moment Jacobian and $\Sigma$ is the asymptotic variance of the empirical moment. The exact NO property of part (i) of Theorem 2 ensures that estimation error in $\eta_{1,k}$ contributes nothing to $\Sigma$; the entire asymptotic variance arises from sampling variation in $K_{\alpha_k}(A-a_k)$ and $\hat{h}(W, a_k, X)$. We note that the $O(N^{-4/5})$ rate in part (iv) of Theorem 2 characterizes the MSE of the $k$-th* moment condition*, not of $\hat{\gamma}$ itself. Under undersmoothing, the moment bias is driven below the $N^{-1/2}$ threshold and $\hat{\gamma}$ achieves the parametric rate, with the anchor-specific bandwidths governing only the rate at which the bias becomes negligible.*

**Remark 1** (Interpretation of the residual $O(\alpha_k^4)$ and the role of local treatment density)**.** *The $O(\alpha_k^4)$ remainder in part (iii) of Theorem 2 merits careful interpretation. It is the error in our* characterization *of the leading bias — the residual of the second-order Taylor expansion — not a second independent source of approximation error. Its magnitude relative to the leading $O(\alpha_k^2)$ bias term is $O(\alpha_k^2)$, which vanishes as $\alpha_k \to 0$, confirming that the leading term dominates asymptotically.*

*Under the optimal anchor-specific bandwidth $\alpha_k^* \propto (N\bar{p}(a_k))^{-1/5}$, where $\bar{p}(a_k) = \mathbb{E}[p(a_k \mid Z, X, W)]$ is the marginal treatment density at $a_k$, both the leading bias and the Taylor remainder acquire concrete $N$-dependent content:*

$$O(\alpha_k^2)\ \propto\ O\Big((N\bar{p}(a_k))^{-2/5}\Big), \qquad O(\alpha_k^4)\ \propto\ O\Big((N\bar{p}(a_k))^{-4/5}\Big). \tag{13}$$

*The remainder is thus of strictly higher order in $N$ than the leading bias, and the approximation improves as the local effective sample size $N\bar{p}(a_k)$ grows. Three implications follow directly.*

*Local density and approximation quality. At anchors where $\bar{p}(a_k)$ is large, the optimal bandwidth $\alpha_k^*$ is small, both the leading bias and the Taylor remainder are small, and the mollifier approximation to the Dirac delta is sharp. In sparse regions $\alpha_k^*$ is larger, but the doubly robust product-of-errors structure — whereby the kernel smoothing bias appears multiplied by the outcome bridge residual $\hat{\varepsilon}$, giving an $O(\alpha_k^2 \cdot \delta_h)$ contribution to the final causal estimate — limits the practical impact on causal inference.*

*Interpretability of the leading bias. The leading bias coefficient*

$$C_k = \tfrac{\kappa_2}{2}\ \mathbb{E}\Big[\partial_a^2 p(a_k \mid Z, X, W)\, q_0\, \hat{h}\Big]$$

*is small precisely when the conditional treatment density is locally flat near $a_k$, the regime in which the bandwidth rule selects small $\alpha_k^*$. The bias is therefore self-limiting in the regions that matter most for causal estimation, and the leading coefficient is identifiable from data and can serve as a diagnostic.*

*Higher-order kernels. For practitioners requiring tighter control of the Taylor remainder, a higher-order kernel satisfying $\int u^j K(u)\, du = 0$ for $j = 1, \ldots, s-1$ pushes the remainder to $O(\alpha_k^{2s})$, making it arbitrarily small relative to the leading bias term without altering the structure of the orthogonalized score $\psi_k^{(1)}$ or the Neyman orthogonality cancellation of part (i).*

**Remark 2** (Structural distinction from Wu et al. [2024])**.** *Theorem 2 differs from Wu et al. [2024] in two fundamental respects.*

*The outcome bridge as test function and doubly robust attenuation. The corrected score $\psi_k^{(1)}$ uses the outcome bridge $\hat{h}(W, a_k, X)$ as the test function, a choice dictated by the proximal Fredholm structure of the treatment bridge equation. The same $\hat{h}$ appears in the doubly robust causal estimator, so the kernel smoothing bias $O(\alpha_k^2)$ in the treatment bridge is attenuated by the outcome bridge residual $\hat{\varepsilon}$ in the final estimate, giving the product-of-errors guarantee $O(\alpha_k^2 \cdot \delta_h)$. This doubly robust attenuation is a direct consequence of the Fredholm structure and is not present in Wu et al. [2024], whose moment conditions do not incorporate the outcome bridge as a test function and therefore do not admit this product form (see Eq. 8 in Wu et al. 2024, where a minimax solution is used and $p(A = a \mid w, x)$ is explicitly modeled).*

*Anchor-specific bandwidth calibration. The bandwidth $\alpha_k$ is anchor-specific, calibrated to the local marginal treatment density $\bar{p}(a_k)$ via $\alpha_k^* \propto (N\bar{p}(a_k))^{-1/5}$. This ensures that the mollifier approximation to the Dirac delta is no coarser than necessary at any anchor, and that the local minimax rate $O(N^{-4/5})$ is achieved uniformly across anchors. A global bandwidth cannot achieve this: in settings where $\bar{p}(a_k)$ varies substantially across anchors — typical of nonuniform treatment distributions in observational studies — a global choice either over-smooths in dense regions or under-smooths in sparse ones. The anchor-specific rule eliminates this tradeoff by treating the bandwidth as a local parameter, consistent with both the local nature of the mollifier approximation and the local density modeling interpretation developed in Section 7.*

# References


Yong Wu, Yanwei Fu, Shouyan Wang, and Xinwei Sun. Doubly robust proximal causal learning for continuous treatments. *Int. Conf. Learning Representations (ICLR)*, 2024.